Statistics 52002 / 52019
Fall 2026 (3 credits)
Course Syllabus

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Course Homepage	`https://github.com/kapelner/HUJI_52002_Fall_2026`
Discord Channel	`https://discord.com/channels/1556341692754567288`
Contact	Write to @kapelner on discord in a public channel
Office	24-417
Lecture Time / Loc	Sun 11:30-14:00 / 21-204
Instructor Office Hours / Loc	see course homepage
TA(s) / Office Hours Time / Loc	Eynat Sela / see course homepage

Course Overview

Big Data Algorithms / Data Mining 3 cr. Prereq.: 52571 (Regression and Statistical Models) OR 52307 (Regression and Linear Models). We plan to cover the following plus/minus other relevant topics:

- working locally and remotely in *nix environments - becoming fluent in *nix commands
- basic git - you are required to have a github account where you will push your assignments
- definition of what big data means (fits in RAM, fits on local disk, does not fit on local disk)
- working with CSV files, compression locally and remotely
- becoming fluent with SQL basics locally and remotely
- columnar formats (Parquet, arrow), duckDB, Polars, data.table
- joins (locally and remotely)
- profiling time/memory
- acquiring data from the web
- data hygiene (deduping, missingness handling)

- basics of hash functions
- streaming and one-pass statistics accuracy-vs-memory tradeoff, error bounds
- sampling statistics: SRS, stratified
- fitting standard models, identify breakdown of standard models (computational and statistical), fitting standard models at scale exactly and approximately (subsampling, bag of little bootstraps)
- statistical significance (pval) vs practical/clinical significance (effect size)
- dimension reduction
- nearest neighbors
- parallel computing with multiple cores and multiple machines
- predictive modeling with tree models, xgboost/lightGBM

We will cover these topics using both R and Python. I plan to use quarto. R is the better language for statistics but Python is the better language for infrastructure. A true data scientist / engineer should be fluent in both. They are not substitutes but different tools. SQL is the substrate for both so we will cover the basics but I don't consider it a "third language". Same as bash scripting - not a "third language" but necessary to become fluent in. As for Julia, scala, Java, Go, SAS, Matlab, etc - some industries use them but they are not widely adopted by data scientists in practice and thus, beyond the scope of the course.

You should be familiar with the following before entering the class:

- Basic Probability Theory: conditional probability, in/dependence, identical distributedness, discrete random variables: Bernoulli, Binomial, expectation and variance
- Linear algebra: Vectors, matrices, transpose, inverse, rank, nullity
- Fundamental programming concepts: basic data types, vectors, arrays, control flow (for, while, if, else), functions
- Basic statistics: point estimation, hypothesis testing, confidence intervals
- Inference vs prediction, theory of modeling

We will review the above *throughout the semester* when needed and we will do so rapidly.

This is not your typical statistics course. This course will do lots of hands-on big data manipulation using modern computing.

Required Technology

Computer: You need your own laptop and it is *required* to bring it to lectures where you can follow the demos on your own machine.

Source Control: You will be expected to use `git` and have a `github.com` account with a class repository (see `01_linux_git_github_setup.qmd`). You will use this repository to submit coding lab assignments (and theory assignments if you use LaTeX). Setting this up is a required course preassignment.

Software

We will be concentrating on the computer language `R` which is a free, open source programming language available for all operating systems. `R` is the most popular language for academic Statistics, Biostatistics and Data Science. Anything cutting-edge in these fields is released in `R` first. However, it is the second-most used language for industry data science, the first being `Python` which we will be using as well. And of course we will be using `SQL` which isn't really a "language" but more of a shorthand for interfacing with databases efficientyl. All these programs and libraries should be set up on your machine. I will have a running software setup documents `02_software_setup_ubuntu.qmd` or `02_software_setup_mac.qmd`, followed by `03_software_setup_common.qmd`.

The 52019 Section

You are the students taking this course as part of a masters degree. Thus, there may be extra homework problems for you and you will be graded on a separate curve.

Announcements

Course announcements will be made via discord in the `#general` channel (not on email). I am known to send a few discord messages per week on important issues.

I can't stress the following enough: if you are not on discord, you will miss all class announcements!!! Discord notifies you when there are messages.

Discord Coupled with Github as a Learning Management System

Each assignment and exam will have its own channel. You can feel free to discuss things with your fellow students there. If you are asking me a question, you must do so in the `#discussions` channel for a general questions or the assignment-specific channel (e.g. `#hw03`) so other students can see the question and benefit from the answer. Do not open "issues" on github! If you pm me for help with a class assignment, I will not answer and just ask you to move it to the appropriate public channel. Do not be afraid to ask questions. There are many people who will have your same question! Discord is to be used professionally so no posting about random stuff!

I will not be using Moodle.

Class Meetings

There are 13 scheduled meetings from Sunday Oct 11 to Sunday Jan 10 (except Dec 6) of which 12 are lectures and one will be the in-person midterm exam (Nov 29).

Lecture format

Lectures are in-person and will be split into two modes: practice and theory. The former is where will be using the "computer/projector" to see the concepts in action and the latter is a standard "chalkboard" lecture where we learn concepts and the second.

Homework

Homework will be split into *theory* and *practice* (called "labs" which are like mini projects). This course is in English and will involve writing *English* and thus being graded in *English*.

Theory Assignments

There will be 3 or 4 theory homework assignments. Homeworks will be assigned and placed on the course homepage and will usually be due a week later. Homework will be **graded** out of 100 with extra credit getting scores possibly > 100. The TA will be doing the grading and will either grade an *arbitrary subset of the assignment* which is determined after the homework is handed in or *will grade based on completeness*.

You will submit your finished homework as a PDF by pushing it to your github repo. To set up your github repo, please follow directions on the course homepage (see the link at the top of the README).

To generate the PDF file of your homework, **you must do one of the following:**

- Download the `hwxx.tex` file and the associated `preamble.tex`, delete the line `\toggletrue{professormode}` to put the file into "student mode". Then write the answers in LaTeX under each problem header.
- **Print out the homework and handwrite your answers in the alotted space for each question. Then scan your homework as a PDF. There are a ton of good photo $\Rightarrow$ PDF apps for both iPhone and droid.**
- **Open the PDF on your device and use a PDF editing program to electronically handwrite your answers and save the PDF.**

I will NOT accept homework that is not atop the original rendered homework PDF file. Remember to write your name.

Lab Assignments

These will almost exclusively consist of short and medium coding exercises in the required software packages of this course (see section above).

Collaboration Policy on all Class Assignments

You are highly recommended to work with each other and help each other. You must, however, submit your own solutions, *with your own write-up* and in *your own words*. There can be no collaboration on the actual *writing*. Failure to comply will result in severe penalties. The university honor code is something I take seriously and I send people to the Dean every semester for violations.

Philosophy of Homework

Homework is the *most* important part of this course. Success comes from experience in working with and thinking about the concepts. It's kind of like weightlifting; you have to lift weights to build muscles. My job as an instructor is to provide assistance through your zone of proximal development. With me, you can grow more than you can alone. To this effect, homework problems are color coded green for easy, yellow for harder, red for challenging and purple for extra credit. You need to know how to do all the greens by yourself. If you've been to class and took notes, they are a joke. Yellows and reds: feel free to work with others. Only do extra credits if you have already finished the assignment. The "[Optional]" problems are for extra practice — highly recommended for exam study.

Time Spent on Homework

This is a 3 credit course. Thus, the amount of work outside of the 2.5hr in-class time per week is 6-7.5hr. I will aim for 6.5hr of homework per week on average. You can think of doing the homework well as "sufficient" as *my* job is to ensure that by *you* doing the homework you will study and understand the concepts in the lectures and you won't have all that much to do when the exams roll around.

Late Assignment Policy

Late homework will be penalized 10 points per day (exept Shabbat) for a maximum of five days. *Do not ask for extensions*; just submit the homework late. After five days, **you can hand it in whenever you want** until *the last scheduled class meeting*.

Reserve Duty If you're called up for reserve duty (or you're the parent of a child under 13 and your spouse is called up) you will not be penalized for lateness. For any assignment due during your service, your deadline is extended by the number of days you served, counted from your release date. Please let me know as soon as you get a call-up. Send me your (or your spouse's) Form 3010 (from your personal area on the reserves website), or an active-service certificate if you're still serving.

Illness For an illness, send me a note from any doctor (your kupat holim doctor is fine) stating the dates you were unable to study. Your deadline will be extended by that many days. Please don't include your diagnosis. I only need the dates, and keeping medical details out protects your privacy. Please don't come to class sick. For chronic conditions or other ongoing needs, contact the Dean of Students so we can set up proper accommodations.

As far as I know, this is one of the most lenient and flexible homework policies in college. I realize things come up. Do not abuse this policy; you will fall behind.

AI Policy

Claude, Claude code, ChatGPT, codex, Gemini, Grok et al. (collectively "AI") is not absolutely required for this course. But we must all acknowledge that it is now an integral (and soon to be ubiquitous) part of the world we live in. Most germane is the fact that sometime in 2024, it reached the point where it can solve the at-home exercises in this course.

On the one hand, this presents a monumental pedagogical challenge. As mentioned above, the exercises are the only way to truly learn the concepts. If you use AI to solve the questions on autopilot, you will never master the concepts. And the exams will subsequently crush you. On the other hand, the educated citizens of the future need to learn to leverage AI to avoid being left behind. But here's my loose policy for those who wish to use AI for the at-home exercises:

- First, try your best to do the HW exercise yourself without it.
- If you can do the exercise, treat the AI as a teaching assistant. Feel free to then show your answer to the AI to see if you missed any steps.
- If you cannot do the exercise after *honestly trying*, treat the AI once again as a teaching assistant who is allowed to help you. Ask it for your missing step and *tell it explicitly not to give you the answer*. Then the next step, and the next, etc. As for its response(s), paraphrase them onto your paper *in your own words* and in the style, notation and dialect we use in this class. This paraphrasing exercise will be effective in helping you learn.
- If you used AI at all for the exercise, be transparent and cite which specify AI model along with date and version (do this once as an endnote at the conclusion of your assignment)
- If you *directly* copy from the AI and you cite it, you will still receive a zero for that exercise. If you directly copy from the AI and you *don't cite the AI*, you will be penalized as this will be considered plagiarism. We can figure out if you copy (as we ourselves use AI daily).

Examinations

Examinations are solely based on homeworks (which are rooted in the lectures)! If you can do all the green and yellow problems on the homeworks, the exams should not present any challenge. I will *never* give you exam problems on concepts which you have not seen at home on one of the weekly homework assignments.

Exams and Major Assignments

- Midterm exam
- Final exam

The schedule of exams and assignment due dates are on the course homepage.

Exam Policies and Materials

I allow you to bring any calculator you wish but it cannot be your phone. The only other items allowed are pencil and eraser. I do not recommend using pen but it is allowed. Food is not allowed during exams but beverages are allowed.

I also allow "cheat sheets" on examinations. For the midterms, you are allowed to bring one 8.5" $\times$ 11" sheet of paper (front and back). Two sheets single sided are not allowed. On this paper you can write anything you would like which you believe will help you on the exam. For

the final, you are allowed to bring two 8.5" × 11" sheet of paper (front and back). Four sheets single sided are not allowed. I will be handing back the mditerm cheat sheets so you can reuse your midterm cheat sheets for the final if you wish.

Cheating on Exams

If I catch you cheating, you can either take a zero on the exam, or you can roll the dice before the University Honor Council who may choose to suspend you.

Missing Exams

There is no midterm exam makeup. But there will be a final exam makeup.

Grading and Grading Policy

Your course grade will be calculated based on the percentages as follows:

Homework (theory and labs)	25%
Midterm Examination[1]	25%
Final Examination (cumulative[2])	40%
Attendance[3]	10%

Checking your grade and class standing

You can always check your grades in real-time using `https://huji.gradesly.com`. You will enter in your ID number (or HUJI email address). I will provide you with your password the first week of class.

Auditing

Auditors are welcome to come to the lectures in-person. They are encouraged to do all course assignments. I will even grade them. Note that the university does not allow auditors to take examinations. Auditors are not allowed to be a "participant" in Moodle as the university policy requires tuition to be paid to be included in Moodle.

[1] If the midterm is missed with a reserve duty 3010 form or doctor's note (see policy in the "Late Assignments" section above), there will be no penalty. Instead, the Final will be worth 65%.

[2] Although the exam is cumulative, it will focus mostly on material after the midterm in order to give a uniform weighting to all course topics when considering the midterm and final weights.

[3] Two free attendance passes are given - so attendance is graded on 10 of the 12 lectures' attendance.